

# MAHMOUD ELLOUMI

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## PROFILE

Computer Science and Applied Mathematics engineering student at ENIS. Across 2 internships and several projects, I have integrated LLMs and generative AI into full-stack applications using Ollama, Mistral AI and n8n. Seeking a **final-year internship (PFE)** to design **AI agents** that solve real business needs.

## EXPERIENCE

- Full-Stack & AI Developer Intern – IDSOFT (ID Software Solutions)** Sfax | 07/2026 – 08/2026  
*AI-enhanced after-sales service (SAV) management app* *React, Spring Boot, PostgreSQL, Ollama*
  - Built an after-sales web app (7-state service-request lifecycle) for 3 roles: customer, technician, manager.
  - Integrated 6 local generative AI features (Ollama, no data sent to third parties): troubleshooting assistant, natural-language data assistant, request classification, best-technician suggestion, report-writing assistance.
  - Designed a layered Spring Boot REST API (Swagger), secured with JWT, TOTP, BCrypt and role-based access control.
  - Created the manager dashboard: interventions by status and priority, revenue, employee performance.
- Mobile Developer Intern – LOGICOM Informatique** Sfax | 07/2025 – 08/2025  
*Product scanning mobile app* *Flutter*
  - Built a Flutter QR-code scanning app covering 40 products, showing available prices and product details.
  - Delivered 3 screens (home, scanner, cart) connected to the company database.

## PROJECTS

- SPARKY – Intelligent Business Document Processing (PFA)** 05/2026  
*Team of 3, Scrum (9 sprints)* *Angular, FastAPI, Mistral AI, PostgreSQL, n8n*
  - Designed the 11-node n8n workflow (Google Sheets export, email notification) and built the Angular 20 frontend: scanner, price comparison, dashboard.
  - AI extraction of invoices, quotes and purchase orders with Mistral AI: 3–8 s per document vs 120–180 s for our n8n + Ollama prototype (90% faster).
  - Compared 3 suppliers across 10 items in under 1 minute, vs about 20 minutes by hand.
  - Containerized all services with Docker Compose; local fallback extraction (pdfplumber + regex) when the AI fails.
- Intelligent Traffic Monitoring – team course project** 2025
  - Real-time vehicle detection and classification on video streams (YOLO, OpenCV, Python), with a dashboard counting vehicles by type.
- TERRANAUT – Drone-Based Detection of Sick Animals – team course project** 2024
  - Applied computer vision and deep learning (object detection) to thermal images captured by drones.

## TECHNICAL SKILLS

|                     |                                                                             |
|---------------------|-----------------------------------------------------------------------------|
| AI & LLMs           | LLMs (Ollama, Mistral AI, Qwen), Generative AI, AI agents, Machine Learning |
| Vision & Automation | Deep Learning, OpenCV, YOLO, PyTorch, OCR, n8n                              |
| Languages           | Python, Java, JavaScript/TypeScript, C                                      |
| Web & Mobile        | React, Angular, Spring Boot, FastAPI, Node.js, Flutter                      |
| Data & DevOps       | PostgreSQL, MySQL, MongoDB, Kafka, Docker, Git, Linux, Scrum                |

## EDUCATION

|                                                                                          |                |
|------------------------------------------------------------------------------------------|----------------|
| Engineering Degree in Computer Science and Applied Mathematics – ENIS, Sfax              | 2024 – Present |
| Preparatory Cycle in Technology – IPEIS, Sfax ( <i>top 100 in the national ranking</i> ) | 2022 – 2024    |
| Baccalaureate in Technical Sciences – 15 November 1955 High School, Sfax                 | 2022           |

## CERTIFICATIONS

NVIDIA Getting Started with Deep Learning • Cisco C Advanced • Scrum (SFC) • Hedera (HCF) • Microsoft MS-900

## VOLUNTEERING (IEEE)

**Photographer** – IEEE ENIS SB (2026) • **Treasurer & Webmaster** – IEEE Cyber Security ENIS (2025)  
**Social Media Manager & Photographer** – TSYP 13 • **Organizer** – Pitch Yourself for a Career

## LANGUAGES & INTERESTS

**Arabic:** native    **French:** upper-intermediate    **English:** intermediate    **Spanish:** elementary  
**Interests:** photography, art, sports, innovation